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Description - This discusses the theory of operation of the TPS, including VME, CPU, AP, and memory.

1- VME INTERFACE BOARD (BIT 3)

There is a VME I/F Board in the TPS/ISE Chassis and another in the host computer. The VME I/F VMEbus-VMEbus Adapter enables a bus master processor on one subsystem to directly address resources on the other subsystem as though they were local. It allows the user the ability to pass blocks of data, input/output (I/O) commands, and interrupts between the two subsystems. The VME I/F Board acts as the lone bus controller in a remote VMEbus chassis, in effect, extending a single VMEbus machine across two backplanes at distances up to fifty feet. Address mapping permits one VMEbus subsystem to directly address the second VMEbus chassis as if it were local memory, via random-access memory reads or writes. The VME I/F Board provides a flexible interface permitting random-access 32-bit transfers to the TPS/ISE subsystem at speeds comparable to read/write speeds on memory cards in the local host computer subsystem.

A DMA controller on each VME I/F Board permits high-speed automatic data transfers from the memory of the host computer subsystem to the memory on the TPS/ISE subsystem. The processor on either subsystem can initiate memory transfers in either direction 64 K to 2 Mb. Longer data blocks may be concatenated in successive DMA cycles. Two page-mode registers combine their contents with the lower sixteen address bits on the VMEbus to permit random access of up to four gigabytes of remote memory through a much smaller address window. VMEbus I/O can be directly addressed by a processor so that reads or writes within an VME I/F Board I/O window are translated into I/O operations on the remote bus.

The two subsystems are linked with a dual-port memory. The address of the dual-port RAM is independently set by jumpers on each VME I/F Board, and may be set to respond to one address range in one subsystem and a different range in the other. Both subsystems can access the memory at the same time and the dual-port RAM arbitrates simultaneous accesses by the two subsystems. Interrupts can pass between the two VMEbus chassis. Any of the seven VMEbus interrupts may be passed to the TPS/ISE subsystem where that system processor can handle the interrupt, and generate an interrupt acknowledge cycle.

The VME I/F Board works by mapping a portion of memory address space in the TPS/ISE subsystem into memory address space on the host computer. The configuration process takes unused memory space in a VMEbus chassis and converts that space to a set of memory which provides transparent access from one backplane to the other.

2- CENTRAL PROCESSING UNIT BOARD (MVME162)

The Central Processing Unit (CPU) consists of a Motorola single board computer that has both a MC68040 microprocessor (µP) and a MC68882 Floating Point Coprocessor (FPC). Both processors are able to run at 25 mHz. There are eight megabytes of 32-bit DRAM (dynamic random-access memory), and a seven-level interrupt handler to service other TPS/ISE boards. A serial RS-232 port is also available for diagnostics. The address bus and the data bus of the CPU Board are configured to accept 32-bit words via on-board jumpers.

The MC68040 is a full 32-bit processor with 32-bit registers, 32-bit data and 32-bit addresses. It is an advanced processor of the MC68000 family.

The MC68882 FPU is designed as a logical extension of the MC68040 processor. It appears as a 32-bit port to µP which runs at the same frequency. It runs the IEEE standard for binary-point arithmetic.

The eight megabytes of DRAM are accessible from the µP, the refresh circuitry, and the VME bus. Request to the DRAM is given by the on-board multi-port arbiter. The arbiter interrogates the status of BR*, BG*, and BGACK* signals on the VME bus to determine ownership of DRAM. The DRAM requires refreshing every eight µsec.

The on-board interrupt handler senses and responds to local interrupts, seven VME bus interrupts, VME bus ACFAIL*, VME bus SYSFAIL*, and the ABORT switch. Depending on the particular interrupt, the µP either generates an on-board interrupt, or a vector branch specific to the device that caused the interrupt.

3- ARITHMETIC PROCESSOR BOARD AND MEMORY

3-1 AP CE

The Arithmetic Processor or AP Board (manufacturing by Mercury Computer Systems, Inc.) performs the calculations that are used in image reconstruction. It is also used for other processes that require large and/or complex calculations to be performed quickly. The AP has one i860 processor that contains nine functional units: the Core Execution Unit, the Floating Point (Control, Adder and Multiplier) Units, Graphics Unit, Paging Unit, Instruction Cache, Data Cache, and Bus and Cache Control Unit. The AP has eight megabytes of internal DRAM with both address and data paths.

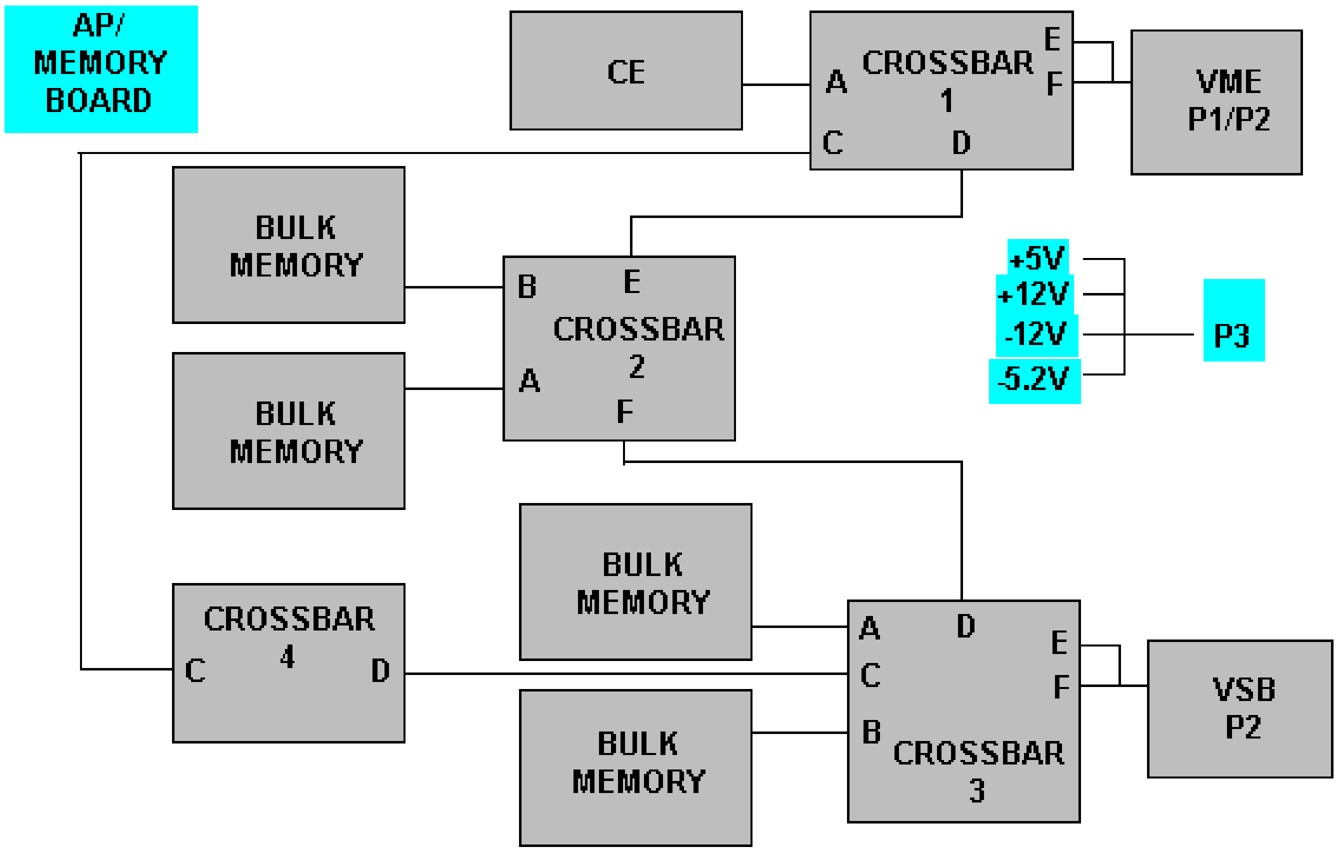
Each functional unit serves a different purpose. The Core Execution Unit controls the overall operation of the i860. This unit handles loads, stores, integer, bit, and program-control operations. The floating point unit controls the floating point adder and multiplier units and graphics unit. This unit handles data transfers, control of the sub-units, and status of the floating point units. The Paging Unit implements protected, paged virtual memory. The Instruction Cache and Data Cache allow for fast instruction fetches and data transfers between either cache and the core execution unit, or the floating point units. The Bus and Cache Control Unit provides an interface between the i860 and the external devices.

The eight megabytes of memory are used to store instruction code, data, the system stack, and the instruction cache. The memory is physically made up of DRAM split into even and odd banks. Dual porting (independent access by two devices) of the memory is used by the host (of the VME bus) and the AP. This allows the host to have direct access to the AP memory. The memory is composed of two physically separate banks of dynamic RAM: one for even addresses, the other for odd addresses. This separation is transparent to software; however, this type of architecture allows for faster access times. For consecutive reads, the access times can reach peak performance of 100nsec (10 mHz). This is equivalent to a maximum memory bandwidth of 80 megabytes/sec.

3-2 AP/Memory Crossbar

The AP/Memory Board uses standard RACE building blocks wherever possible (this includes RACE CE modules, crossbar ASICs, and the VME interface). The AP/Memory Board mothercard is a VME-compatible 9U x 400 mm printed circuit card.

The architecture of the basic AP/Memory Board consists of multiple crossbar ASICs interconnecting a VME bus interface, a VSB bus interface, connectors for up to three standard, two processor (or I/O) daughtercards (supported by six crossbar connections), and connectors for up to four bulk memory (128 Mb) daughtercards. For low end systems, standard 32-Mb memory daughtercards (large memory CEs with the i860 removed) are used as the bulk memory card. Due to differences in form factor, a 32-Mb memory card occupies two bulk memory locations (but uses only one crossbar port). See Illustration 1 for the functional block diagram. See section 3-3 for descriptions and theory.



AP/MEMORY
ILLUSTRATION 1

3-3 Functional Descriptions

AP CE

The AP/Memory Board uses standard RACE building blocks wherever possible (this includes RACE CE modules, Crossbar ASICs, and the VME interface). The AP/Memory Board mothercard is a VME compatible 9U x 400 mm printed circuit card.

The architecture of the basic AP/Memory Board consists of multiple crossbar ASICs interconnecting a VME bus interface, a VSB bus interface, connectors for up to three standard, two processor (or I/O) daughtercards (supported by six crossbar connections), and connectors for up to four bulk memory (128 Mb) daughtercards. For low end systems, standard 32-Mby memory daughtercards (large memory CEs with the i860 removed) are used as the bulk memory card. Due to differences in form factor, a 32-Mby memory card occupies two bulk memory locations (but uses only one crossbar port).

AP/Memory Crossbar

Each crossbar supports three concurrent transfer paths. Any transfer in the system may take place concurrently with any other as long as the two transfers do not use the same port on any crossbar in the path. Conflicts are resolved by the hardware.

Transfers requested by the VSB slave interface have priority through the crossbars. A crossbar transfer generated by VSB master accessing the bulk memory causes any other crossbar transfers to suspend until the VSB/crossbar access is complete. Since the VSB interface contains a FIFO, crossbar transfers use the bandwidth efficiently. Because of their lower priority, VME transfers to /from the bulk memory can be suspended for extended periods by VSB accesses. For this reason, the VME bus controller should have a bus timeout value greater than 20 milliseconds to prevent access conflicts from causing VME bus timeouts.

P/Memory VME Bus

The VME interface is a full-functioned master/slave interface.

Bulk Memory

The Bulk Memory cards are DRAM daughtercards with characteristics as described below. The Bulk Memory card interface is a standard crossbar port.

128 Mb Bulk Memory Features

- 128 Megabytes of memory.
- 32-bit parity (1 parity bit for 4 bytes of data).
- DMA controller programmable from any crossbar master.

32 Mb Bulk Memory Features

- 32 Megabytes of memory.
- 4 parity bits for 4 bytes of data (same memory requirements as byte parity).
- DMA controller programmable from any crossbar master.

AP/Memory VSB Bus

The VSB interface is a master/slave interface designed to a subset of the VSB specification. Only those features required for high performance data transfers are supported by the interface.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 2, 1998	J. Saperstein	Initial Conversion from Toolbook to Word.
1	Oct 13, 1999	M. Keber	Added correct proprietary heading to document.